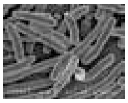
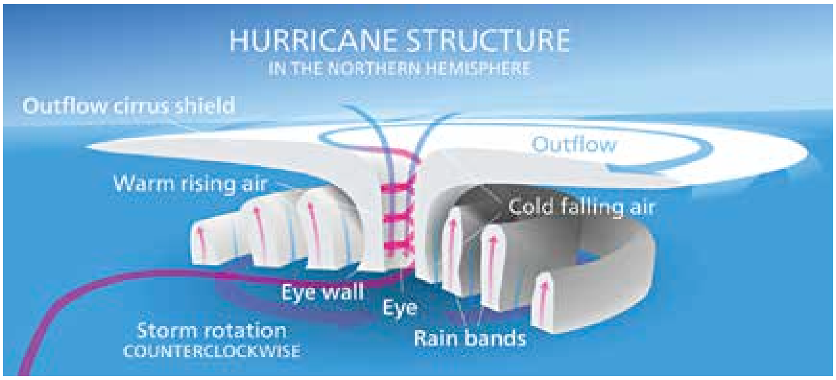


Killing bacteria with light
Scientists have synthesized a molecule which is converted into an antibiotic by bacteria themselves when illuminated. **http://dgit.in/LightKill**

Repetitive sounds = music
Even the most mundane sounds when repeated over and over again can appear melodious to the brain. **http://dgit.in/BaDumTis**

Hurricane Harvey as on August 24. 2017



The typical structure of a tropical cyclone in the Northern Hemisphere

impress you, look again. Hurricane Harvey is the costliest tropical storm on record, causing more than $200bil-lion in damages. It also broke a 12 year streak of no major hurricanes making landfall in the United States. That's not all – with peak rainfall measurements of 1,539mm Harvey was also the wettest tropical cyclone on record in the US.

HURRICANE SEASONS AROUND THE WORLD

Season Name	Starts	Ends
Atlantic Hurricane Season	June 1	November 30
Eastern Pacific Hurricane Season	May 15	November 30
Northwest Pacific Typhoon Season	all year	all year
North Indian Cyclone Season	April 1	December 31
Southwest Indian Cyclone Season	October 15	May 31
Australian/Southeast Indian Cyclone Season	October 15	May 31
Australian/Southwest Pacific Cyclone Season	November 1	April 30

HURRICANE OPHELIA (2017)

Just like Harvey, if you're looking for higher wind speeds and lower pressures on Ophelia, you'd be disap-pointed. The significance of Ophelia lies elsewhere. Tropical storms usu-ally form on warm waters – which is conducive to create the conditions required (temperature difference, pres-sure drop) for wind speed to increase. That is why it came as a surprise to meteorologists and ordinary people alike that Hurricane Ophelia affected Ireland, the UK and parts of Europe. It was the easternmost Atlantic major hurricane on record.

If you want to grasp just how unexpected the storm was, take a look at the image that the US National Hurricane Centre tweeted (below). It didn't account for storms going north of 60 degrees North latitude or eastward than 0 degrees longitude – ending up with a jarringly incom-plete storm prediction graph. Ophelia wasn't just a surprise this year – it was the strongest storm to hit Ireland since the 1960s.

All of these big and unique storms bring us to one question – why are storms defying reason and where are they headed?

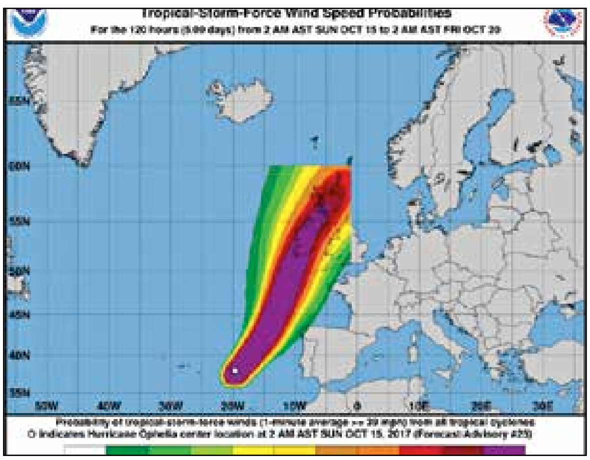


Image Source: Offaly Express

The NHC chart that shows the incomplete prediction.

CLIMATE CHANGE?

While there has been much debate about whether climate change is real or not, there is one undeniable fact about the issue that we can confidently share with you – IT IS REAL! Now with that taken care of, it isn't really straight-forward to blame cli-mate change for the increasing frequen-cy, range and strength of storms. And no, as stated earlier, we're not being climate change naysayers here.

When it comes to a storm forma-tion, there is a wide array of factors responsible – some might say too wide. But some of these factors do point at bigger storms. For instance, warmer ocean waters and higher sea levels are both favorable to increas-ing the intensity of storms – and both are directly affected by climate change. While warmer seas directly affect wind speed, both the factors contribute to increased flooding – warm waters causing increased precipitation and higher sea level causing more coastal flooding.

Even though it might appear that storms are increasing in frequency, that is not entirely true especially if you look at historic storm season data. Some models even project a decrease in frequency. What is hap-pening, though, is that whatever storms do occur now, have higher intensities. And if climate change keeps heading in the direction it is right now, there's no sign of Earth's stormy days and nights being over anytime soon. d